CardDemo statement generation: wiki pages and I/O contract

Wiki pages generated by the FlowX.AI legacy transformation agent from AWS CardDemo (open source, Apache-2.0). CONTRACT.md is the input/output contract for the build.

Pages included:

- `02-processes/process-statement-generation.md`
- `10-acceptance-criteria/acceptance-criteria-statement-generation.md`
- `01-data-model/project-data-model.md`
- `04-integrations/workflow-account-data-access.md`
- `04-integrations/workflow-batch-file-transfer.md`
- `04-integrations/workflow-transaction-data-access.md`
- `07-plugins/plugin-statement-document.md`
- `02-processes/process-end-of-day-batch-chain.md`
- `CONTRACT.md`

`02-processes/process-statement-generation.md`

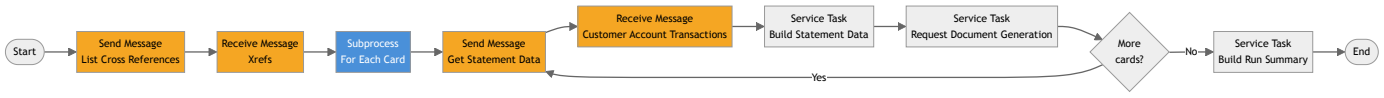
Process: statement-generation

Property	Value
Type	Main Process (batch)
Legacy Source	Job `CREASTMT` (`/app/jcl/CREASTMT.JCL`), programs `CBSTM03A` (driver) and `CBSTM03B` (file I/O subroutine), copybooks `COSTM01`, `CVACT01Y`, `CVACT03Y`, `CVCUS01Y`, `CVTRA05Y`
Trigger	Called by end-of-day-batch-chain
Roles	Batch Operator (system); statement recipients are customers
Integration	workflow-account-data-access, workflow-transaction-data-access, workflow-batch-file-transfer
Plugin	plugin-statement-document
Acceptance Criteria	acceptance-criteria-statement-generation
Mapping	P-17

Purpose

Produces one statement per card/account with customer name, address, balances, FICO score and the list of the account's transactions, in text (80 columns) and HTML (100 columns).

Process Flow



Business Rules

#	Rule	Legacy Source
BR-01	Statement driven by CardXref order (card number)	`CBSTM03A`
BR-02	Transactions are keyed by card number + transaction id (sorted copy `TRXFL.SEQ` loaded into KSDS `TRXFL.VSAM.KSDS`, key length 32)	`CREASTMT.JCL`
BR-03	Statement lines: transaction id, description, amount; total expenditure (`Total EXP`) = sum of the charges listed (amounts above zero: purchases, interest); credits and returns are listed with their amounts but not subtracted	`CBSTM03A` {{TODO: confirm line layout from COSTM01 / STMT-LINES}}
BR-04	Output text 80 columns (STMTFILE), HTML 100 columns (HTMLFILE)	`CBSTM03A`
BR-05	The DD for STMTFILE in `CREASTMT.JCL` is corrupted (finding F-14) — not migrated	`CREASTMT.JCL`
BR-06	The legacy driver uses ALTER/GO TO, a 2-D array and control-block addressing; FlowX replaces it with a loop subprocess	`CBSTM03A`

Data Mapping

SEND `GET_STATEMENT_DATA` {cardNumber} → RECEIVE `{xref, account, customer, transactions[]}`. Document request: template `statement-html` and `statement-text`, payload = `StatementData` → output

`statementDocumentRef`.

Configuration Dependencies

Document template ids (`statement-html`, `statement-text`) — Configuration Parameters; statement period {{TODO: legacy prints all transactions on file; confirm cycle filter requirement}}.

Acceptance Criteria: statement-generation

Process: statement-generation (P-17) · **UI Flow:** none (batch) · **Plugin:** plugin-statement-document · **Legacy:** `CBSTM03A/B` / `CREASTMT`

Scenarios

#	Scenario	Steps	Expected Result	Tags
1	Statement per card	1. Run with transactions	One statement per card with customer name, address, account id, balance, FICO score	@smoke @regression
2	Transaction lines	1. Card with 3 transactions	3 lines, total equals the sum of the amounts above zero	@regression
3	Two formats	1. Run	Text and HTML statements both available	@regression
4	Card without transactions	1. Run	Statement with zero total	@regression
5	Delivery	1. Run	Statement stored for the customer `{{DECISION: delivery channel — see notification-strategy}}`	@manual
6	PDF statement	1. Request PDF	Not available in legacy (FI-09)	@manual
7	Credits on the statement	1. Card with purchases 100.00 and 40.00 and a return of –30.00	3 lines; total 140.00 (the return is listed, not subtracted)	@regression

▶ Gherkin

Legacy Comparison

Behaviour	Legacy	FlowX	Same?
Program structure	ALTER / GO TO, in-memory table	Process Definition + document plugin	Redesigned
Output	STMTFILE and HTMLFILE	document template (text + HTML)	Same

Configuration Dependencies

Document template `StatementTemplate`; `StmtFile` DD is corrupted in the legacy JCL (F-14) — recreate from the program.

Data Mapping Verification

StatementData built from Customer, Account, Card, CardXref, Transactions; verify the total and the address formatting.

Project Data Model

This page defines every **Project Data Type** of the AWS CardDemo FlowX application, derived from the legacy VSAM record layouts (`/app/cpy/*.cpy`), the optional-module IMS segments (`CIPAUSMY`, `CIPAUDTY`) and the DB2 DDL (`AUTHFRDS`, `TRANSACTION_TYPE`, `TRANSACTION_TYPE_CATEGORY`). The legacy-to-FlowX mapping is in the Transformation Matrix.

> All legacy VSAM / IMS / DB2 stores are replaced by **FlowX Persistence (App Data Store)**. Access always goes through an Integration Workflow (DATA_SOURCE node on the `carddemo-datastore` NOSQL_DB System). {{DECISION: Keep one NOSQL_DB System (`carddemo-datastore`) with one collection per Project Data Type, instead of separate systems per legacy store (VSAM / IMS / DB2).}}

Data Type Index

ID	Project Data Type	Legacy Source	Legacy Store / Record Size	Collection
DT-01	Account	`CVACT01Y`	VSAM KSDS ACCTDAT / 300	`accounts`
DT-02	Customer	`CVCUS01Y`	VSAM KSDS CUSTDAT / 500	`customers`
DT-03	Card	`CVACT02Y`	VSAM KSDS CARDDAT (+AIX CARDAIX) / 150	`cards`
DT-04	CardXref	`CVACT03Y`	VSAM KSDS CCXREF (+AIX CXACAIX) / 50	`card_xrefs`
DT-05	Transaction	`CVTRA05Y`	VSAM KSDS TRANSACT / 350	`transactions`
DT-06	TransactionCategoryBalance	`CVTRA01Y`	VSAM KSDS TCATBALF / 50	`tran_cat_balances`
DT-07	DisclosureGroup	`CVTRA02Y`	VSAM KSDS DISCGRP / 50	`disclosure_groups`
DT-08	TransactionType	`CVTRA03Y`, DB2 `TRANSACTION_TYPE`	VSAM TRANTYPE / 60, DB2	`transaction_types`
DT-09	TransactionCategory	`CVTRA04Y`, DB2 `TRANSACTION_TYPE_CATEGORY`	VSAM TRANCATG / 60, DB2	`transaction_categories`
DT-10	UserAccount	`CSUSR01Y`	VSAM KSDS USRSEC / 80	`user_accounts`
DT-11	DailyTransaction	`CVTRA06Y`	Sequential DALYTRAN / 350	`daily_transactions`
DT-12	RejectedTransaction	`CBTRN02C` (DALYREJS)	Sequential / 430	`rejected_transactions`

ID	Project Data Type	Legacy Source	Legacy Store / Record Size	Collection
DT-13	PendingAuthSummary	`CIPAUSMY` (IMS root)	IMS PAUTSUM0	`pending_auth_summaries`
DT-14	PendingAuthDetail	`CIPAUDTY` (IMS child)	IMS PAUTDTL1	`pending_auth_details`
DT-15	AuthFraudRecord	DB2 `CARDDEMO.AUTHFRDS`	DB2 table	`auth_fraud_records`
DT-16	SessionContext	`COCOM01Y` (COMMAREA)	CICS COMMAREA	process data (`session`)
DT-17	StatementData	`CBSTM03A/B` (in-memory)	Working storage + STMTFILE/HTMLFILE	`statements`
DT-18	BranchExportRecord	`CVEXPORT` / `CBEXPORT`	Sequential EXPFILE	`export_records`
DT-19	OptionalModuleToggles	CICS `INQUIRE PROGRAM` check (COMEN01C)	None	FeatureToggle Data Type

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DT-01 Account

Legacy source: `/app/cpy/CVACT01Y.cpy` (300 bytes). Primary key `accountId`.

Attribute	Type	Required	Legacy Source	Description
accountId	String(11) numeric	Yes	`ACCT-ID` 9(11)	Account number, non-zero
activeStatus	Enumeration `AccountActiveStatus`	Yes	`ACCT-ACTIVE-STATUS` X(1)	Y / N
currentBalance	Number (2 dec)	Yes	`ACCT-CURR-BAL` S9(10)V99	Current balance
creditLimit	Number (2 dec)	Yes	`ACCT-CREDIT-LIMIT` S9(10)V99	Credit limit
cashCreditLimit	Number (2 dec)	Yes	`ACCT-CASH-CREDIT-LIMIT` S9(10)V99	Cash credit limit
openDate	Date (YYYY-MM-DD)	Yes	`ACCT-OPEN-DATE` X(10)	Account open date
expiryDate	Date	Yes	`ACCT-EXPIRAION-DATE` X(10) (legacy typo)	Account expiry date
reissueDate	Date	No	`ACCT-REISSUE-DATE` X(10)	Reissue date
currentCycleCredit	Number (2 dec)	Yes	`ACCT-CURR-CYC-CREDIT` S9(10)V99	Credits in current cycle
currentCycleDebit	Number (2 dec)	Yes	`ACCT-CURR-CYC-DEBIT` S9(10)V99	Debits in current cycle
addressZip	String(10)	No	`ACCT-ADDR-ZIP` X(10)	Billing ZIP
groupId	String(10)	Yes	`ACCT-GROUP-ID` X(10)	Disclosure Group id (A000000000 / DEFAULT / ZEROAPR)

Legacy FILLER (178 bytes) is dropped. Misspelling `EXPIRAION` is corrected (see Feature Improvements).

DT-02 Customer

Legacy source: `/app/cpy/CVCUS01Y.cpy` (500 bytes). Primary key `customerId`.

Attribute	Type	Required	Legacy Source	Description
customerId	String(9) numeric	Yes	`CUST-ID` 9(9)	Customer id
firstName	String(25)	Yes	`CUST-FIRST-NAME` X(25)	Alphabetic + spaces only
middleName	String(25)	No	`CUST-MIDDLE-NAME` X(25)	
lastName	String(25)	Yes	`CUST-LAST-NAME` X(25)	
addressLine1..3	String(50)	Line 1,2: Yes	`CUST-ADDR-LINE-1..3` X(50)	Street, line 2, city/other
addressState	String(2)	Yes	`CUST-ADDR-STATE-CD` X(2)	US state code
addressCountry	String(3)	Yes	`CUST-ADDR-COUNTRY-CD` X(3)	Country code
addressZip	String(10)	Yes	`CUST-ADDR-ZIP` X(10)	
phone1, phone2	String(15)	Phone1: Yes	`CUST-PHONE-NUM-1/2` X(15)	Format (AAA)BBB-CCCC
ssn	String(9) numeric (**sensitive**)	Yes	`CUST-SSN` 9(9)	Stored/displayed in clear in legacy; encrypt in FlowX
governmentId	String(20)	No	`CUST-GOVT-ISSUED-ID` X(20)	
dateOfBirth	Date	Yes	`CUST-DOB-YYYY-MM-DD` X(10)	Validated not in the future
eftAccountId	String(10)	No	`CUST-EFT-ACCOUNT-ID` X(10)	
primaryCardHolder	Boolean	Yes	`CUST-PRI-CARD-HOLDER-IND` X(1)	Y/N
ficoScore	Integer (300-850)	No	`CUST-FICO-CREDIT-SCORE` 9(3)	

DT-03 Card

Legacy source: `/app/cpy/CVACT02Y.cpy` (150 bytes). Primary key `cardNumber`; secondary index on `accountId` (replaces AIX `CARDAIX`).

Attribute	Type	Required	Legacy Source	Description
cardNumber	String(16) numeric	Yes	`CARD-NUM` X(16)	
accountId	String(11)	Yes	`CARD-ACCT-ID` 9(11)	FK to Account
cvv	String(3) (**sensitive**)	Yes	`CARD-CVV-CD` 9(3)	Clear text in legacy; tokenise
embossedName	String(50)	Yes	`CARD-EMBOSSED-NAME` X(50)	Alphabetic + spaces only
expiryDate	Date	Yes	`CARD-EXPIRAION-DATE` X(10)	Month 1-12, year validated

Attribute	Type	Required	Legacy Source	Description
activeStatus	Enumeration `CardActiveStatus`	Yes	`CARD-ACTIVE-STATUS` X(1)	Y / N

DT-04 CardXref

Legacy source: `/app/cpy/CVACT03Y.cpy` (50 bytes). Primary key `cardNumber`; secondary index on `accountId` (replaces AIX `CXACAIX`).

Attribute	Type	Required	Legacy Source	Description
cardNumber	String(16)	Yes	`XREF-CARD-NUM` X(16)	
customerId	String(9)	Yes	`XREF-CUST-ID` 9(9)	FK to Customer
accountId	String(11)	Yes	`XREF-ACCT-ID` 9(11)	FK to Account

DT-05 Transaction

Legacy source: `/app/cpy/CVTRA05Y.cpy` (350 bytes). Primary key `transactionId` (16 chars). Secondary access by `cardNumber` + processed timestamp (legacy AIX/`TRANIDX` — see {{TODO: confirm alternate index definition in /app/jcl/TRANIDX.jcl}}).

Attribute	Type	Required	Legacy Source	Description
transactionId	String(16)	Yes	`TRAN-ID` X(16)	Legacy: last record + 1 (collision risk); FlowX: sequence generator
typeCode	Enumeration `TransactionTypeCode`	Yes	`TRAN-TYPE-CD` X(2)	01-07
categoryCode	Enumeration `TransactionCategoryCode`	Yes	`TRAN-CAT-CD` 9(4)	
source	String(10)	Yes	`TRAN-SOURCE` X(10)	e.g. POS TERM, OPERATOR
description	String(100)	Yes	`TRAN-DESC` X(100)	
amount	Number (2 dec)	Yes	`TRAN-AMT` S9(9)V99	Format -99999999.99
merchantId	String(9) numeric	Yes	`TRAN-MERCHANT-ID` 9(9)	
merchantName	String(50)	Yes	`TRAN-MERCHANT-NAME` X(50)	
merchantCity	String(50)	Yes	`TRAN-MERCHANT-CITY` X(50)	
merchantZip	String(10)	Yes	`TRAN-MERCHANT-ZIP` X(10)	

Attribute	Type	Required	Legacy Source	Description
cardNumber	String(16)	Yes	`TRAN-CARD-NUM` X(16)	FK to Card
originalTimestamp	DateTime	Yes	`TRAN-ORIG-TS` X(26)	
processedTimestamp	DateTime	Yes	`TRAN-PROC-TS` X(26)	

DT-06 TransactionCategoryBalance

Legacy source `CVTRA01Y` (50 bytes). Composite key `accountId + typeCode + categoryCode`.

Attribute	Type	Required	Legacy Source	Description
accountId	String(11)	Yes	`TRANCAT-ACCT-ID` 9(11)	
typeCode	String(2)	Yes	`TRANCAT-TYPE-CD` X(2)	
categoryCode	String(4)	Yes	`TRANCAT-CD` 9(4)	
balance	Number (2 dec)	Yes	`TRAN-CAT-BAL` S9(9)V99	Running balance per category; basis of interest calculation

DT-07 DisclosureGroup

Legacy source `CVTRA02Y` (50 bytes). Composite key `groupId + typeCode + categoryCode`. Seed data: `/app/data/ASCII/discgrp.txt` (51 rows; groups `A000000000`, `DEFAULT`, `ZEROAPR`).

Attribute	Type	Required	Legacy Source	Description
groupId	String(10)	Yes	`DIS-ACCT-GROUP-ID` X(10)	
typeCode	String(2)	Yes	`DIS-TRAN-TYPE-CD` X(2)	
categoryCode	String(4)	Yes	`DIS-TRAN-CAT-CD` 9(4)	
interestRate	Number (2 dec)	Yes	`DIS-INT-RATE` S9(4)V99	Annual percent; seed file stores signed overpunch (`0150{` = 15.00) — convert at load

DT-08 TransactionType

Legacy sources `CVTRA03Y` / DB2 `TRANSACTION_TYPE(TR_TYPE CHAR(2) PK, TR_DESCRIPTION VARCHAR(50))` / `trantype.txt`.

Attribute	Type	Required	Legacy Source	Description
typeCode	String(2)	Yes	`TRAN-TYPE` / `TR_TYPE`	PK
description	String(50)	Yes	`TRAN-TYPE-DESC` / `TR_DESCRIPTION`	

DT-09 TransactionCategory

Legacy sources `CVTRA04Y` / DB2 `TRANSACTION_TYPE_CATEGORY` (PK `TRC_TYPE_CODE`+`TRC_TYPE_CATEGORY`, FK to TRANSACTION_TYPE ON DELETE RESTRICT) / `trancatg.txt` (18 rows).

Attribute	Type	Required	Legacy Source	Description
typeCode	String(2)	Yes	`TRAN-TYPE-CD` / `TRC_TYPE_CODE`	FK to TransactionType; delete restricted when categories exist
categoryCode	String(4)	Yes	`TRAN-CAT-CD` / `TRC_TYPE_CATEGORY`	
description	String(50)	Yes	`TRAN-CAT-TYPE-DESC` / `TRC_CAT_DATA`	

DT-10 UserAccount

Legacy source `/app/cpy/CSUSR01Y.cpy` (80 bytes), VSAM KSDS USRSEC key (8,0). Sample users in `DUSRSECJ.jcl`.

Attribute	Type	Required	Legacy Source	Description
userId	String(8)	Yes	`SEC-USR-ID` X(8)	Upper-cased on sign-on
firstName	String(20)	Yes	`SEC-USR-FNAME` X(20)	
lastName	String(20)	Yes	`SEC-USR-LNAME` X(20)	
password	Credential hash (**sensitive**)	Yes	`SEC-USR-PWD` X(8)	Plain text in legacy; replaced by FlowX IAM credential {{DECISION: Authentication delegated to the FlowX identity provider; `password` is not stored in the Project Data Type.}}
userType	Enumeration `UserType`	Yes	`SEC-USR-TYPE` X(1)	A = Admin, U = Regular

DT-11 DailyTransaction

Legacy `CVTRA06Y` (350 bytes), sequential file DALYTRAN (input of POSTTRAN). Same attributes as Transaction with prefix `DALYTRAN-` plus `cardNumber` used for xref lookup; additionally `batchId` and `receivedTimestamp` (new, FlowX-only).

DT-12 RejectedTransaction

Legacy: DALYREJS output of `CBTRN02C` = 350-byte transaction image + validation trailer.

Attribute	Type	Required	Legacy Source	Description
dailyTransaction	Object (DailyTransaction)	Yes	DALYREJS record body	Original record

Attribute	Type	Required	Legacy Source	Description
rejectReason	Enumeration `RejectReasonCode`	Yes	`WS-VALIDATION-FAIL-REASON` 9(4)	100-103
rejectReasonDescription	String(76)	Yes	`WS-VALIDATION-FAIL-REASON-DESC`	{{TODO: confirm trailer layout (4 + 76) in CBTRN02C working storage}}
reviewStatus	String	No	New	OPEN / RESOLVED (see Rejected Transaction view)

DT-13 PendingAuthSummary (optional authorization module)

Legacy `CIPAUSMY` (IMS root segment).

Attribute	Type	Required	Legacy Source	Description
accountId	String(11)	Yes	`PA-ACCT-ID` S9(11) COMP-3	Key
customerId	String(9)	Yes	`PA-CUST-ID` 9(9)	
authStatus	String(1)	No	`PA-AUTH-STATUS` X	
accountStatus	List of String(2) (max 5)	No	`PA-ACCOUNT-STATUS` X(2) x5	
creditLimit	Number	Yes	`PA-CREDIT-LIMIT` S9(9)V99 COMP-3	
cashLimit	Number	Yes	`PA-CASH-LIMIT`	
creditBalance	Number	Yes	`PA-CREDIT-BALANCE`	
cashBalance	Number	Yes	`PA-CASH-BALANCE`	
approvedCount / declinedCount	Integer	Yes	`PA-APPROVED-AUTH-CNT` / `PA-DECLINED-AUTH-CNT` S9(4) COMP	
approvedAmount / declinedAmount	Number	Yes	`PA-APPROVED-AUTH-AMT` / `PA-DECLINED-AUTH-AMT`	

DT-14 PendingAuthDetail (optional authorization module)

Legacy `CIPAUDTY` (IMS child segment). Key: `accountId` + `authDate` + `authTime` (stored inverted in IMS for descending order).

Attribute	Type	Required	Legacy Source	Description
accountId	String(11)	Yes	parent key	FK to PendingAuthSummary
authDate / authTime	Date / Time	Yes	`PA-AUTHORIZATION-KEY` (COMP-3)	
originalDate / originalTime	Date / Time	Yes	`PA-ORIG-DATE/TIME`	
cardNumber	String(16)	Yes	`PA-CARD-NUM` X(16)	
authType	String(4)	Yes	`PA-AUTH-TYPE` X(4)	
cardExpiry	String(4)	Yes	`PA-CARD-EXPIRY-DATE` X(4)	MMYY

Attribute	Type	Required	Legacy Source	Description
messageType / messageSource	String(6)	Yes	`PA-MESSAGE-TYPE`, `PA-MESSAGE-SOURCE`	
authIdCode	String(6)	Yes	`PA-AUTH-ID-CODE`	
responseCode	Enumeration `AuthResponseCode`	Yes	`PA-AUTH-RESP-CODE` X(2)	88 APPROVED = '00'
responseReason	Enumeration `AuthDeclineReasonCode`	Yes	`PA-AUTH-RESP-REASON` X(4)	
processingCode	Integer(6)	Yes	`PA-PROCESSING-CODE` 9(6)	
transactionAmount / approvedAmount	Number	Yes	`PA-TRANSACTION-AMT`, `PA-APPROVED-AMT` S9(10)V99	
merchantCategoryCode	String(4)	Yes	`PA-MERCHANT-CATAGORY-CODE`	
acquirerCountryCode	String(3)	No	`PA-ACQR-COUNTRY-CODE`	
posEntryMode	Integer(2)	No	`PA-POS-ENTRY-MODE` 9(2)	
merchantId / merchantName / merchantCity / merchantState / merchantZip	String	Yes	`PA-MERCHANT-ID` X(15), `-NAME` X(22), `-CITY` X(13), `-STATE` X(2), `-ZIP` X(9)	
transactionId	String(15)	Yes	`PA-TRANSACTION-ID`	
matchStatus	Enumeration `AuthMatchStatus`	Yes	`PA-MATCH-STATUS` X(1)	P/D/E/M
fraudFlag	Enumeration `AuthFraudFlag`	No	`PA-AUTH-FRAUD` X(1)	F/R
fraudReportDate	Date	No	`PA-FRAUD-RPT-DATE` X(8)	

DT-15 AuthFraudRecord (optional authorization module)

Legacy DB2 `CARDDEMO.AUTHFRDS` (written by `COPAUS2C`). PK `cardNumber + authTimestamp`. Attributes mirror PendingAuthDetail plus `accountId` (DECIMAL 11) and `customerId` (DECIMAL 9), plus fraud action (`F` confirm / `R` remove) and `fraudReportDate`. Legacy source column: each attribute maps 1:1 to the like-named `AUTHFRDS` column in `/app/app-authorization-ims-db2-mq/ddl/AUTHFRDS.ddl`.

DT-16 SessionContext (COMMAREA replacement)

Legacy `COCOM01Y` (CARDDEMO-COMMAREA). In FlowX this is **process data**, not a persisted type; it is passed from the Sign-on process to Menu and onwards via Call Activity input/output mapping.

Attribute	Type	Legacy Source	Description
fromTransactionId / fromProgram	String	`CDEMO-FROM-TRANID`, `CDEMO-FROM-PROGRAM`	Navigation origin (replaced by FlowX

Attribute	Type	Legacy Source	Description
			navigation)
toTransactionId / toProgram	String	`CDEMO-TO-TRANID`, `CDEMO-TO-PROGRAM`	Navigation target
userId	String(8)	`CDEMO-USER-ID`	
userType	Enumeration `UserType`	`CDEMO-USER-TYPE` (88 CDEMO-USRTYP-ADMIN/USER)	
programContext	Enumeration `ProgramContext`	`CDEMO-PGM-CONTEXT` (88 ENTER/REENTER)	Not needed in FlowX (stateful process)
customerId / accountId / cardNumber	String	`CDEMO-CUST-ID`, `CDEMO-ACCT-ID`, `CDEMO-CARD-NUM`	Selected record context
customerFirst/Middle/LastName	String	`CDEMO-CUST-FNAME/MNAME/LNAME`	
lastMap / lastMapset	String	`CDEMO-LAST-MAP`, `CDEMO-LAST-MAPSET`	Not needed

DT-17 StatementData

Built in memory by `CBSTM03A`/`CBSTM03B` (xref → customer → account → transactions, using 2-D table, ALTER/GO TO control flow). Output formats: fixed text `STMTFILE` and HTML `HTMLFILE`.

Attribute	Type	Legacy Source	Description
statementId	String	New	
accountId, customerId, cardNumber	String	xref	
customerName, customerAddress	String	Customer	
currentBalance, creditLimit	Number	Account	
ficoScore	Integer	Customer	Printed on statement
lines	List of Transaction summaries	Transaction file	id, description, amount
totalExpenditure	Number	computed	Sum of the lines with an amount above zero
statementDocumentRef	String	New	Reference to Document Plugin output

DT-18 BranchExportRecord

`CBEXPORT` writes a multi-record-type export (customer, account, xref, transaction, card) into a single sequential file (`CVEXPORT` layout); `CBIMPORT` reads it back and splits into target files with an error file. Record-type codes (from `CBEXPORT`): `C` = customer, `A` = account, `X` = card cross-reference, `T` = transaction, `D` = card. Common header: record type (1), timestamp (26), sequence number (COMP), branch id (4), region code (5), then a 460-byte data area redefined per type (total 500 bytes). Note the export copies `ssn` in clear (finding in Legacy Overview).

DT-19 OptionalModuleToggles (FeatureToggle Data Type)

Replaces the CICS `INQUIRE PROGRAM` check used by `COMEN01C` to hide menu options of non-installed optional modules. See Enumerations and the config bucket rules in Architecture Overview.

Attribute	Type	Default	Description
pendingAuthorizationEnabled	Boolean	false	Menu option 11 (COPAUS0C)
transactionTypeMaintenanceEnabled	Boolean	false	Admin options 5 and 6 (COTRTLIC / COTRTUPC)
mqInquiryEnabled	Boolean	false	MQ date/account inquiry services

Relationships

See Data Model Diagram. Key referential rules (legacy enforced in code, not in VSAM):

- Card → Account via `accountId`; Card → Customer via CardXref.
- Transaction → Card; TransactionCategoryBalance → Account + TransactionCategory.
- TransactionCategory → TransactionType (DB2 FK ON DELETE RESTRICT; VSAM had none).
- Account.groupId → DisclosureGroup.groupId (fallback `DEFAULT` when no row for the group).

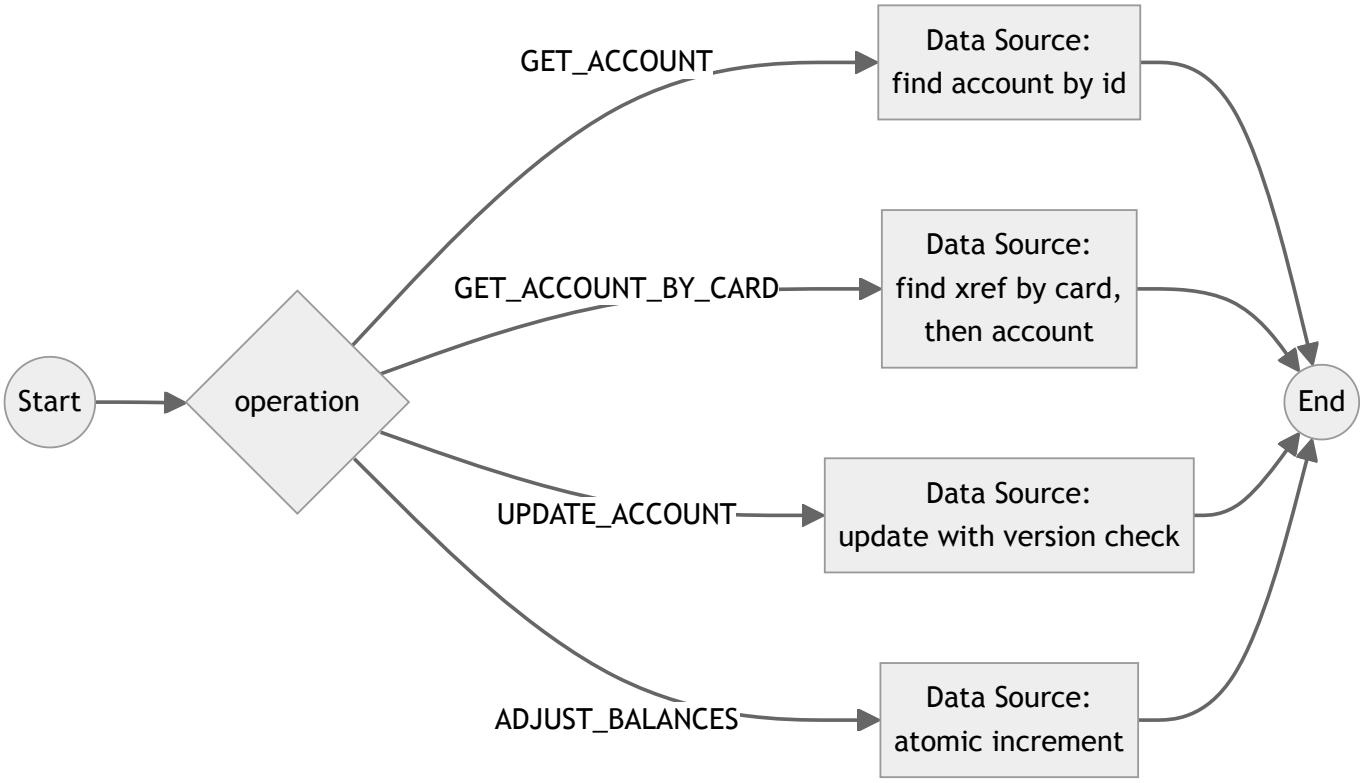
Workflow: account-data-access

Metadata

Field	Value
System	carddemo-datastore (NOSQL_DB)
Trigger Source	Process (MESSAGE_EVENT_SEND with StartIntegrationWorkflowHandler)
Trigger Processes	P-03, P-04, P-10, P-11, P-14, P-15, P-17, P-22, P-28
Correlation	`processInstanceId`
Legacy Source	`READ/REWRITE ACCTDAT`, `CARDXREF` alternate-index reads in `COACTVWC`, `COACTUPC`, `COBIL00C`, `COTRN02C`, `CBTRN02C`, `CBACT04C`, `COACCT01`
Mapping	W-01

{{DECISION: One workflow per Project Data Type with an `operation` input and a Condition node that routes to the matching DATA_SOURCE node, to keep the number of workflows manageable; split into one workflow per operation if independent versioning is preferred.}}

Workflow Flow



Nodes and Operations

Operation	Node	Data Source operation	Input	Output
GET_ACCOUNT	find account	find one in `accounts` by `accountId`	`accountId`	`account`, `found`
GET_ACCOUNT_BY_CARD	find xref, find account	find one in `cardXrefs` by `cardNumber`, then `accounts`	`cardNumber`	`account`, `accountId`, `customerId`, `found`
UPDATE_ACCOUNT	update account	update `accounts` where `accountId` and `version` = `expectedVersion`	`account`, `expectedVersion`	`updated`, `conflict`
ADJUST_BALANCES	adjust	increment `currentBalance`, `cycleCredit`, `cycleDebit`	`accountId`, deltas	`account`, `status`

Data Mappings

SEND payload (process → workflow): `operation`, `accountId` / `cardNumber`, `account` (for update), `expectedVersion`, deltas. RECEIVE payload (workflow → process, written to `accountResult`): `found`, `conflict`, `account.*`, `status`, `message`.

Direction	Source	Target
Process → Send	`${selectedAccountId}`	`jsonDTOMessage.accountId`
Workflow → API	`workflow.accountId`	filter `accountId`
API → Workflow	`result.document`	`workflow.account`
End → Receive	`END.payload.account`	`accountResult.account`

Error Handling

Error	Handling	Outcome
Not found (legacy `RESP NOTFND`)	`found = false`	Process shows the "not found" message
Version conflict	`conflict = true`	Process shows "Record changed by some one else. Please review"
Store timeout	retry 3 times with backoff	then error status to process, no data change

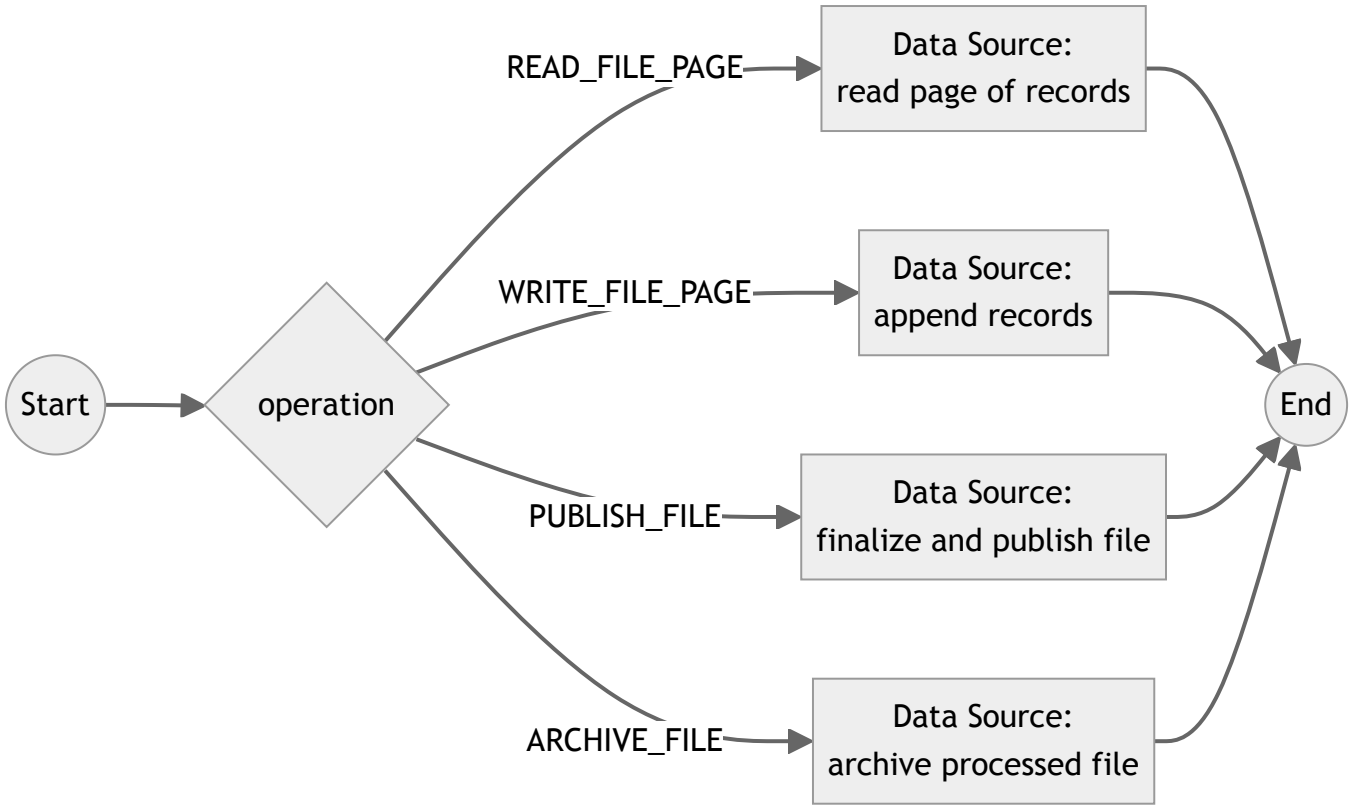
Workflow: batch-file-transfer

Metadata

Field	Value
System	batch-file-gateway (Kafka custom connector)
Trigger Source	Process (MESSAGE_EVENT_SEND with StartIntegrationWorkflowHandler)
Trigger Processes	P-14, P-15, P-16, P-17, P-18, P-20, P-21, P-26
Legacy Source	JCL `DD` statements and `REPRO`/`IDCAMS` steps in `/app/jcl/`
Mapping	W-09

{{DECISION: Files are read in pages (cursor) and returned as structured records instead of streaming the complete dataset through one message, because Kafka messages are size-limited.}}

Workflow Flow



Operations

Operation	Input	Output
READ_FILE_PAGE	`fileName`, `cursor`, `pageSize`	`records[]`, `nextCursor`, `recordCount`
WRITE_FILE_PAGE	`fileName`, `records[]`, `sequence`	`written`

Operation	Input	Output
PUBLISH_FILE	`fileName`, `format` (text, HTML, fixed-width)	`fileReference`, `size`
ARCHIVE_FILE	`fileName`	`archivedReference`

Data Mappings

RECEIVE (`fileResult`): `records[]`, `nextCursor`, `fileReference`, `status`. Dataset names (legacy `AWS.M2.CARDDEMO.*`) become logical file names; `datasetHighLevelQualifier` is a Configuration Parameter.

Error Handling

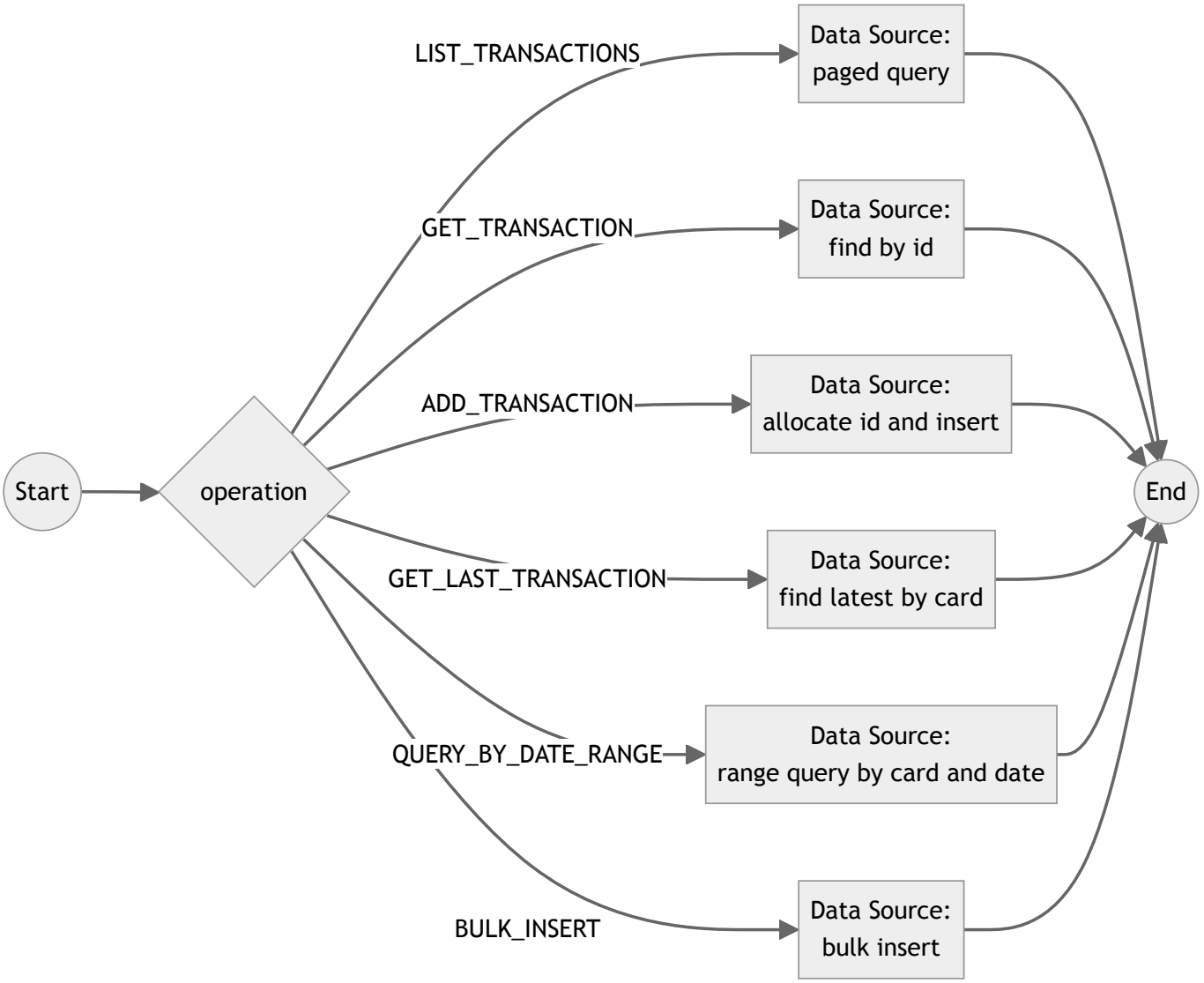
File not found / empty → status `NO_DATA`; read error → retry 3 times, then error status. Legacy file status codes (`00`, `10`, `35`, `39`) are mapped to `OK`, `END`, `NOT_FOUND`, `FORMAT_ERROR` {{TODO: confirm file status handling per program}}.

Workflow: transaction-data-access

Metadata

Field	Value
System	carddemo-datastore (NOSQL_DB)
Trigger Source	Process (MESSAGE_EVENT_SEND with StartIntegrationWorkflowHandler)
Trigger Processes	P-08, P-09, P-10, P-11, P-14, P-15, P-16, P-17, P-18
Legacy Source	`COTRN00C` – `COTRN02C`, `COBIL00C` (`TRANSACT` browse/read/write), `CBTRN02C`, `CBACT04C`, `CBTRN03C`, `CBSTM03A`, `COMBTRAN.jcl`
Mapping	W-04

Workflow Flow



Nodes and Operations

Operation	Data Source operation	Input	Output
LIST_TRANSACTIONS	query `transactions` ordered by id, start at `startTransactionId`, page of 10	`startTransactionId`, `direction`, `pageSize`	`transactions[]`, `hasNext`, `hasPrevious`
GET_TRANSACTION	find by `transactionId`	`transactionId`	`transaction`, `found`
ADD_TRANSACTION	insert; the new id is allocated by a sequence (replaces "last record + 1")	`transaction`	`transactionId`
GET_LAST_TRANSACTION	latest transaction of a card (PF5 copy-last)	`cardNumber`	`transaction`
QUERY_BY_DATE_RANGE	transactions between `startDate` and `endDate` (report, statement)	`cardNumber/accountId`, dates, `cursor`	`transactions[]`, `nextCursor`
BULK_INSERT	insert many (posting, interest, load)	`transactions[]`	counts

Data Mappings

RECEIVE (`transactionResult`): `transactions[]` / `transaction.*`, `found`, `transactionId`, counts, `nextCursor`. The id sequence replaces the `STARTBR/READPREV/ENDBR` last-record logic (F-10 duplicate id collision).

Error Handling

Not found → `found = false`; duplicate id → retry with a new sequence value; timeout → retry 3 times then error status; bulk insert failures are returned per record.

Plugin: statement-document

Property	Value
Plugin Type	Document Plugin (template-based document generation)
Used by	statement-generation (P-17)
Legacy Source	`CBSTM03A`, `CBSTM03B`, copybooks `COSTM01`, `CVACT01Y`, `CVCUS01Y`, `CVTRA05Y`; job `CREASTMT.JCL`
Mapping	PL-01

Content

Section	Data	Legacy Source
Header	statement date, account id	`CBSTM03A`
Customer block	name, address (3 lines, state, zip), phone	`CVCUS01Y`
Account summary	credit limit, current balance, cycle credit/debit, FICO score	`CVACT01Y`, `CVCUS01Y`
Transaction lines	transaction id, description, amount per card of the account	`CVTRA05Y`
Totals	total expenses (sum of the amounts above zero; credits and returns are not subtracted)	`CBSTM03A`

Data Mapping

Template variable	Source (process data)
`statement.accountId`	`statementData.accountId`
`statement.customer.*`	`statementData.customer.*`
`statement.transactions[]`	`statementData.transactions[]`
`statement.totalExpenses`	`statementData.totalExpenses`

Formats: text (80 columns) and HTML as in legacy; PDF new. SSN is not printed. {{TODO: confirm the exact statement line layouts in CBSTM03A (not analyzed in detail).}}

Process: end-of-day-batch-chain

Property	Value
Type	Main Process (orchestrator for batch Call Activities)
Legacy Source	Control-M folders (`/app/scheduler/CardDemo.controlm`), CA-7 definition (`/app/scheduler/CardDemo.ca7`, schedule 000652), jobs `CLOSEFIL`, `TRANBKP`, `WAITSTEP`, `OPENFIL`, `POSTTRAN`, `INTCALC`, `COMBTRAN`, `CREASTMT`, `TRANREPT`, `DISCGRP`; assembler `MVSWAIT` / COBOL `COBSWAIT`
Trigger	Scheduled Trigger (daily / weekly / monthly) — see Trigger Inventory
Roles	Batch Operator (system)
Acceptance Criteria	acceptance-criteria-end-of-day-batch-chain
Mapping	P-19

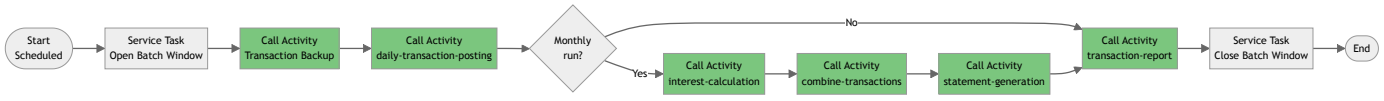
Purpose

Coordinates the nightly/monthly batch cycle that in the mainframe is: close CICS files (`CLOSEFIL`) → post transactions → interest → combine → statements → reports → reopen CICS files (`OPENFIL`).

{{DECISION: In FlowX there are no CICS files to close/open. `CLOSEFIL`, `OPENFIL` and `WAITSTEP` (pure wait) are **dropped** from the technical flow and replaced by a processing-window flag (`batchWindowOpen`) that blocks online data-changing processes during the run.}}

{{TODO: confirm with business whether online access must be blocked during the batch run}}

Process Flow



Legacy Job Mapping

Legacy Job	Program	Control-M Folder	FlowX Node
CLOSEFIL	IEFBR14/CICS close	DAILY, WEEKLY(DISCGRP), MONTHLY	Open Batch Window
TRANBKP	IDCAMS REPRO	DAILY-TransactionBackup	Transaction Backup (Call Activity: export `transactions` snapshot)
WAITSTEP	`COBSWAIT` / `MVSWAIT`	DAILY, WEEKLY, MONTHLY	dropped
OPENFIL	CICS open	DAILY, WEEKLY, MONTHLY	Close Batch Window
POSTTRAN	`CBTRN02C`	(chain)	daily-transaction-posting

Legacy Job	Program	Control-M Folder	FlowX Node
INTCALC	`CBACT04C`	MONTHLY-InterestCalculation	interest-calculation
COMBTRAN	SORT/REPRO	MONTHLY-InterestCalculation	combine-transactions
CREASTMT	`CBSTM03A/B`	(chain)	statement-generation
TRANREPT	`CBTRN03C`	(chain)	transaction-report
DISCGRP	IDCAMS REPRO	WEEKLY-DisclosureGroupsRefresh	master-data-load (reference refresh)
MNTTRDB2	`COBTUPDT`	WEEKLY-TransactionTypesDBRefresh	transaction-type-batch-maintenance
CBPAUP0J	`CBPAUP0C`	triggered from CLOSEFIL (CA-7 SCHID 030)	expired-authorization-purge

Scheduler Facts

- Control-M folders (APPLICATION `CARD-DEMO`, RUN_AS `xuemingh`, TIMETO 23:00): DAILY-TransactionBackup, WEEKLY-TransactionTypesDBRefresh, WEEKLY-DisclosureGroupsRefresh, MONTHLY-InterestCalculation.
- `TRANEXTR` is listed under the WEEKLY-DisclosureGroupsRefresh parent folder (misplacement, finding F-16).
- CA-7: system `CARDDEMO`, JCLLIB `&CARDDEMOPRODJCL`, schedule 000652; `CLOSEFIL` triggers `CBPAUP0J` (SCHID=030).
- Cron expressions are not defined in the extracted definitions: {{TODO: obtain run days/times (the Control-M XML shows all months enabled and TIMETO=23:00 only)}}.

Business Rules

#	Rule	Legacy Source
BR-01	Order: posting before interest, interest before combine, combine before statements, statements before reports	Control-M / README job chain
BR-02	A failing step stops the chain (`COND=(0,NE)` on later steps)	`CREASTMT.JCL`, others
BR-03	Posting with rejects (`RETURN-CODE 4`) does not stop the chain	`CBTRN02C`
BR-04	Run window ends by 23:00 (`TIMETO`)	Control-M

Data Mapping

Process start data: `batchId`, `businessDate`, `runType` (DAILY / WEEKLY / MONTHLY). Call Activity INPUT: `batchId`, `businessDate`, `interestRunDate`. OUTPUT: `status`, `processedCount`, `rejectedCount`.

Configuration Dependencies

`schedulerEndTime`, `interestRunDate`, `reportStartDate`, `reportEndDate`, `batchWindowOpen` (runtime flag → FeatureToggle Data Type `OptionalModuleToggles` extension {{DECISION: add `batchWindowOpen` to a

separate `RuntimeSwitches` FeatureToggle}}).

Candidate I/O contract

What a migrated implementation must produce so the scorer can compare it with the oracle. The contract fixes **formats only**. It says nothing about behaviour, so it may be handed to a builder without leaking the answers.

Inputs (given per run)

The same files the COBOL chain reads, one record per line, fixed width, layouts as in the copybooks:

File	Copybook	Record
`dailytran.dat`	CVTRA06Y	350
`acctdata.dat`	CVACT01Y	300
`cardxref.dat`	CVACT03Y	50
`tcatbal.dat`	CVTRA01Y	50
`discgrp.dat`	CVTRA02Y	50
`custdata.dat`, `carddata.dat`, `trantype.dat`, `trancatg.dat`	CUSTREC, CVACT02Y, CVTRA03Y, CVTRA04Y	500, 150, 60, 60

Run parameters: business date `2022-07-18`, INTCALC parameter `2022071800`.

Step isolation. A candidate for INTCALC receives the oracle's POSTTRAN outputs as its account and category-balance inputs, so an INTCALC result is never penalised for a POSTTRAN error.

Outputs

One JSON file per output: a list of objects, in any order. Field names and types are fixed. Strings may omit trailing spaces. Amounts are JSON strings or numbers holding at most two decimals (`"14.56"`, `-500`).

File	Step	Fields
`posttran_transact.json`	POSTTRAN	transaction fields (below) for every posted transaction
`posttran_dalyrejs.json`	POSTTRAN	transaction fields + `reason` (int) + `reason_desc` for every rejected transaction
`posttran_acctdata.json`	POSTTRAN	account fields for every account, after posting
`posttran_tcatbal.json`	POSTTRAN	`acct_id, type_cd, cat_cd, bal` for every category balance, after posting
`intcalc_systran.json`	INTCALC	transaction fields for every interest transaction written
`intcalc_acctdata.json`	INTCALC	account fields for every account, after interest

- Transaction fields: `tran_id, type_cd, cat_cd (int), source, desc, amt, merchant_id (int), merchant_name, merchant_city, merchant_zip, card_num, orig_ts, proc_ts`
- Account fields: `acct_id (int), active_status, curr_bal, credit_limit, cash_credit_limit, open_date, expiration_date, reissue_date, curr_cyc_credit, curr_cyc_debit, addr_zip, group_id`

A run may instead leave the oracle's fixed-width dumps (`<name>.txt`). The scorer reads either form.

Optional `flags.json`: the uncertainties the pipeline declared **before** scoring, as `[{"rule": "R-INT-TRUNC", "source": "<artefact path and line>"}]` or `[{"output": "...", "key": "...", "source": "..."}]`. See PROTOCOL.md §6 for what counts as a flag.

Candidate I/O contract, layer 1 supplement (statement generation, daily transaction report)

This supplements CONTRACT.md for the two layer 1 steps. Like CONTRACT.md it fixes **formats only**: which files a run receives, which files it writes, and where each field sits on a printed line. It says nothing about how a field's value is computed or assembled (which records go into a statement or a total, how a name or an address line is built, which records are selected). That is behaviour, and it comes from the wiki.

Inputs (given per run)

Fixed-width text, one record per line, layouts as in the copybooks (CONTRACT.md).

Step	File	Copybook	Record
statement generation	`transact.dat`: the day's combined transaction file	CVTRA05Y	350
statement generation	`acctdata.dat`: accounts after the day's posting and interest	CVACT01Y	300
statement generation	`cardxref.dat`	CVACT03Y	50
statement generation	`custdata.dat`	CUSTREC	500
daily transaction report	`transact.dat`: the day's combined transaction file	CVTRA05Y	350
daily transaction report	`cardxref.dat`	CVACT03Y	50
daily transaction report	`trantype.dat`, `trancatg.dat`	CVTRA03Y, CVTRA04Y	60, 60

Run parameters: daily transaction report start date `2022-07-01` and end date `2022-07-31` (`YYYY-MM-DD`).

Step isolation. Each step receives the oracle's upstream outputs (the combined transaction file and the accounts after interest), so a result is never penalised for an upstream error.

Outputs

Each output is a text file of fixed-length lines, one line per printed line, in the order printed. Trailing spaces may be omitted. Instead of a file, a run may return the same lines as one string (lines joined by `\n`) or as a list of strings, under the file's name without extension.

File	Step	Line length
`creastmt_statements.txt`	statement generation	80
`creastmt_statements_html.txt`	statement generation	100
`tranrept_report.txt`	daily transaction report	133

Optional `assumptions.md` and `flags.json` as in CONTRACT.md.

Picture notation in the tables below, as in COBOL: `X(n)` text of n characters, left-aligned, space-padded; `9(9).99-` nine digits with leading zeros, a point, two decimals and a trailing sign (`-` or a space); `Z(9).99-` the same with leading zeros printed as spaces; `+ZZZ,ZZZ,ZZZ.ZZ` / `-ZZZ,ZZZ,ZZZ.ZZ` a sign in the first position (`+`/`-`, or space/`-`), leading zeros and their commas printed as spaces.

`creastmt_statements.txt`: one statement, lines in this order

#	Line (positions are 1-based)
0	`*` × 31, `START OF STATEMENT`, `*` × 31
1	1–75 customer name X(75); 76–80 spaces
2	1–50 address line 1 X(50); 51–80 spaces
3	1–50 address line 2 X(50); 51–80 spaces
4	1–80 address line 3 X(80)
5	`-` × 80
6	33 spaces, `Basic Details`, spaces to 80
5	`-` × 80
7	1–20 `Account ID          :`; 21–40 account ID X(20); spaces to 80
8	1–20 `Current Balance     :`; 21–33 current balance `9(9).99-`; spaces to 80
9	1–20 `FICO Score          :`; 21–40 FICO score X(20); spaces to 80
10	`-` × 80
11	30 spaces, `TRANSACTION SUMMARY`, spaces to 80
12	`-` × 80
13	`Tran ID` padded to 16, `Tran Details` padded to 51, `  Tran Amount` (80 in all)
12	`-` × 80
14	one per transaction: 1–16 transaction ID X(16); 17 space; 18–66 transaction details X(49); 67 `$`; 68–80 amount `Z(9).99-`
12	`-` × 80
14A	1–10 `Total EXP:`; 11–66 spaces; 67 `$`; 68–80 total `Z(9).99-`
15	`*` × 32, `END OF STATEMENT`, `*` × 32

`creastmt_statements_html.txt`: one statement, lines in this order

Each line is one of the fixed lines below or a fixed line with one field inserted (`{…}`), padded to 100. The transaction block repeats once per transaction; then the closing block ends the statement.

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>HTML Table Layout</title>
</head>
<body style="margin:0px;">
<table  align="center" frame="box" style="width:70%; font:12px Segoe UI,sans-serif;">
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#1d1d96b3;">
<h3>Statement for Account Number: {account ID X(20)}</h3>
```

```
</td>
</tr>
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#FFAF33;">
<p style="font-size:16px">Bank of XYZ</p>
<p>410 Terry Ave N</p>
<p>Seattle WA 99999</p>
</td>
</tr>
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#f2f2f2;">
<p style="font-size:16px">{customer name}  </p>
<p>{address line 1}  </p>
<p>{address line 2}  </p>
<p>{address line 3}  </p>
</td>
</tr>
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#33FFD1; text-align:center;">
<p style="font-size:16px">Basic Details</p>
</td>
</tr>
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#f2f2f2;">
<p>Account ID         : {account ID X(20)}</p>
<p>Current Balance    : {current balance 9(9).99-}</p>
<p>FICO Score         : {FICO score X(20)}</p>
</td>
</tr>
<tr>
<td colspan="3" style="padding:0px 5px;background-color:#33FFD1; text-align:center;">
<p style="font-size:16px">Transaction Summary</p>
</td>
</tr>
<tr>
<td style="width:25%; padding:0px 5px; background-color:#33FF5E; text-align:left;">
<p style="font-size:16px">Tran ID</p>
</td>
<td style="width:55%; padding:0px 5px; background-color:#33FF5E; text-align:left;">
<p style="font-size:16px">Tran Details</p>
</td>
<td style="width:20%; padding:0px 5px; background-color:#33FF5E; text-align:right;">
<p style="font-size:16px">Amount</p>
</td>
</tr>
--- transaction block, once per transaction ---
<tr>
<td style="width:25%; padding:0px 5px; background-color:#f2f2f2; text-align:left;">
<p>{transaction ID X(16)}</p>
</td>
<td style="width:55%; padding:0px 5px; background-color:#f2f2f2; text-align:left;">
<p>{transaction details X(49)}</p>
</td>
<td style="width:20%; padding:0px 5px; background-color:#f2f2f2; text-align:right;">
<p>{amount Z(9).99-}</p>
</td>
</tr>
--- closing block ---
<tr>
```

```
<td colspan="3" style="padding:0px 5px;background-color:#1d1d96b3;">
<h3>End of Statement</h3>
</td>
</tr>
</table>
</body>
</html>
```

`tranrept_report.txt` : line types

Line	Layout (positions are 1-based)
report header	1–38 `DALYREPT` ; 39–79 `Daily Transaction Report` ; 80–91 `Date Range: ` ; 92–101 start date; 102–105 `to` ; 106–115 end date
blank line	spaces
column header	`Transaction ID` padded to 17, `Account ID` to 12, `Transaction Type` to 19, `Tran Category` to 35, `Tran Source` to 14, one space, `Amount` padded to 16
rule	`-` × 133
detail	1–16 transaction ID; 17 space; 18–28 account ID; 29 space; 30–31 type code; 32 `-` ; 33–47 type description X(15); 48 space; 49–52 category code 9(4); 53 `-` ; 54–82 category description X(29); 83 space; 84–93 source X(10); 94–97 spaces; 98–112 amount `-ZZZ,ZZZ,ZZZ.ZZ` ; 113–114 spaces
page total	1–11 `Page Total` ; 12–97 `.` × 86; 98–112 total `+ZZZ,ZZZ,ZZZ.ZZ`
account total	1–13 `Account Total` ; 14–97 `.` × 84; 98–112 total `+ZZZ,ZZZ,ZZZ.ZZ`
grand total	1–11 `Grand Total` ; 12–97 `.` × 86; 98–112 total `+ZZZ,ZZZ,ZZZ.ZZ`

A page starts with the report header, the blank line, the column header and a rule. Each page total and each account total line is followed by a rule.